

Psychoacoustic Metrics for the Drone-Noise Pipeline

Math report: SQM and Psychoacoustic Annoyance

Source: Vaiuso, Righi, Coretti, Apicella (ZHAW, 2024) — arXiv:2410.22208

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Abstract

This is the math we'd need to wire a **Psychoacoustics** tab into SoundVisualizer. Four sound-quality metrics (SQM) — *loudness*, *sharpness*, *roughness*, *fluctuation strength* — and the composite *Psychoacoustic Annoyance* (PA) index that combines them. The framework predates this paper; the paper applies it to drone propeller noise specifically. All metrics rest on the **Bark** / **critical-band** model of hearing (24 bands across the audible range).

1 Loudness L (sone)

How loud the sound is perceived to be. Reference: 1 sone = a pure 1 kHz tone at 40 dB SPL. Standardized as ISO 532-1 / DIN 45631 (Zwicker method).

$$L = \int_{z=0}^{24} L'(z) dz$$

$L'(z)$ *specific loudness* as a function of critical-band rate z in Bark (units: sone/Bark). Computed per Bark band from the SPL via a non-linear power law plus inter-band masking corrections (the ISO 532B algorithm).

z critical-band rate, $0 \leq z \leq 24$ Bark. Maps non-linearly to frequency: 1 Bark $\approx$ 100 Hz at low frequencies, scaling up to $\approx$ 20 kHz at 24 Bark.

Two flavours matter for us:

- **Stationary loudness** (ISO 532B / DIN 45631) — one number per recording.
- **Time-varying loudness** $L(t)$ (DIN 45631/A) — computed in ~ 2 ms steps. Required for PA, which uses the 5th-percentile N_5 (i.e. "loudness exceeded 5 % of the time" — captures peaks, not steady-state).

2 Sharpness S (acum)

How shrill or piercing the sound is. Reference: 1 acum = narrow-band noise centered at 1 kHz, 60 dB, 160 Hz bandwidth. Paper uses the DIN 45692 weighting (a refined Aures / Von Bismarck form).

$$S = \frac{k}{L} \int_{z=0}^{24} L'(z) g(z) z dz$$

$g(z)$ weighting function emphasizing high critical-band rates (the high-Bark end of the spectrum).

k method-dependent scaling constant; calibrated so that the reference signal evaluates to exactly 1 acum.

L total loudness from the previous section.

Geometrically: S is the *loudness-weighted centroid of the Bark axis*, biased toward the high end by $g(z)$. Sounds with more energy at high frequencies have higher sharpness.

3 Roughness R (asper)

How rough / harsh the sound is. Caused by amplitude modulation at 30–300 Hz — fast fluctuations the ear cannot track individually. Reference: 1 asper = 60 dB 1 kHz tone, 100 % modulated at 70 Hz.

$$R = \text{cal} \cdot \int_{z=0}^{24} f_{\text{mod}} \Delta L \, dz$$

f_{mod} modulation frequency per Bark band (Hz), located by peak-picking the envelope spectrum.

ΔL modulation depth per band: $\max(\text{env}) - \min(\text{env})$ in dB.

cal calibration constant so the reference signal evaluates to 1 asper.

Drone noise tends to score *high* on roughness because the blade-pass frequency and its harmonics modulate each other in the audible 30–300 Hz envelope range.

4 Fluctuation Strength F (vacil)

How wobbly the sound is. Driven by *slow* amplitude modulation, 0–30 Hz, with peak sensitivity at 4 Hz. Reference: 1 vacil = 60 dB 1 kHz tone, 100 % modulated at 4 Hz.

$$F = \frac{0.008 \cdot \int_{z=0}^{24} \Delta L \, dz}{\frac{f_{\text{mod}}}{4} + \frac{4}{f_{\text{mod}}}}$$

Same structural form as roughness, but the denominator $f_{\text{mod}}/4 + 4/f_{\text{mod}}$ has its *minimum* (and therefore F its maximum) at $f_{\text{mod}} = 4$ Hz — matching the empirical ear-sensitivity peak.

For drones, F is typically near zero (~ 0.01 vacil mean in the paper's hover measurements) — rotor modulations sit well above the slow-fluctuation regime.

5 Psychoacoustic Annoyance PA

The composite that ties the four SQMs into one number. Zwicker & Fastl form:

$$PA = N_5 \left(1 + \sqrt{w_S^2 + w_{FR}^2} \right)$$

where the loudness peak N_5 is multiplied by a unitless "unpleasantness factor" $(1 + \sqrt{w_S^2 + w_{FR}^2})$. The two weights are:

Sharpness weight w_S :

$$w_S = \begin{cases} (S - 1.75) \cdot 0.25 \cdot \log_{10}(N_5 + 10), & \text{if } S > 1.75 \\ 0, & \text{if } S \leq 1.75 \end{cases}$$

Below 1.75 acum, sharpness does not contribute. Above the threshold the contribution grows with both the sharpness excess and the overall loudness.

Roughness + fluctuation weight w_{FR} :

$$w_{FR} = \frac{2.18}{N_5^{0.4}} (0.4 F + 0.6 R)$$

Roughness gets 1.5× the weight of fluctuation strength. The $N_5^{-0.4}$ factor softens the impact of these terms for already-loud sounds.

Mental model. $PA \approx \text{loudness} \times \text{timbre-unpleasantness multiplier}$. Loudness dominates; sharpness and roughness modulate.

6 Observed values for drones

The paper reports the following for a DJI Matrice 300 RTK in steady hover:

Metric	Mean	Max	Unit
Loudness (L)	19.30	29.56	sone
Sharpness (S)	1.85	3.69	acum
Roughness (R)	0.28	0.95	asper
Fluctuation (F)	0.01	0.06	vacil
Annoyance (PA)	20.31	32.57	—

Observations:

- Sharpness sits *just* above the 1.75 acum activation threshold — the w_S term contributes weakly.
- Fluctuation strength is essentially noise-floor and contributes almost nothing to w_{FR} .
- Roughness is the dominant timbre contributor through the $0.6 \cdot R$ term.
- Therefore PA for a hovering drone is approximately $L \times 1.05$ — the timbre weights bump it up by a few percent above raw loudness.

7 Implementation path

The paper computed these with MATLAB's Audio Toolbox (proprietary). Open-source Python options:

Library	What it provides	Trade-off
mosquito	All four SQM + PA, ISO/DIN-compliant, BSD-licensed, pip-installable	Adds ~ 10 MB dependency, but the canonical choice
scikit-maad, pyAudioAnalysis	Loudness and some SQM	Less complete; PA not built in
Roll our own	Full control + insight	Weeks of work; non-trivial filter banks and ISO 532B masking model

Recommendation: add `mosquito` to `pyproject.toml`. Server-side, compute the five numbers once per acoustic measurement and cache them alongside `meta.json`. Frontend reads cached numbers; no computation in the browser.

Suggested Psychoacoustics tab views.

1. **Per-mic table at the selected PWM point** — one row per mic with L , S , R , F , PA columns. Same shape as the FFT-tab header strip but per-mic.
2. **Metrics vs. elevation** — small-multiple line plots of each metric across elevation. Surfaces the spatial-radiation pattern of the unpleasantness components, not just total loudness.
3. **Metrics vs. PWM (or RPM, or thrust)** — mirror of the Custom-tab scatter, with any SQM or PA on the Y axis. Answers questions like "at what RPM does this prop start sounding annoying for reasons other than loudness?"

Open questions for the implementation phase.

- **Calibration.** Sone / acum / asper / vacil all assume the input is in dB SPL, not dBFS. We need the UMIK-2 calibration applied *before* feeding audio to `mosquito`. Already the case if a cal file is uploaded for the mic; otherwise the absolute values will be off (relative comparisons still work).
- **Time window.** For time-varying loudness $\rightarrow N_5$, we need $\gtrsim 1$ s of audio. Our typical capture is 1.5 s per PWM step; marginal but workable.
- **Norsonic comparison.** Once the NOR-145 lands, its built-in loudness/sharpness reading is a sanity check against our `mosquito` pipeline.